SUPERNOVA × SPACEPOINT · THEME 03 · SEAS OF DATA

Radar sees every ship. We make each one checkable.



WORKING PROTOTYPE

REAL SENTINEL-1 DATA

LIVE DASHBOARD

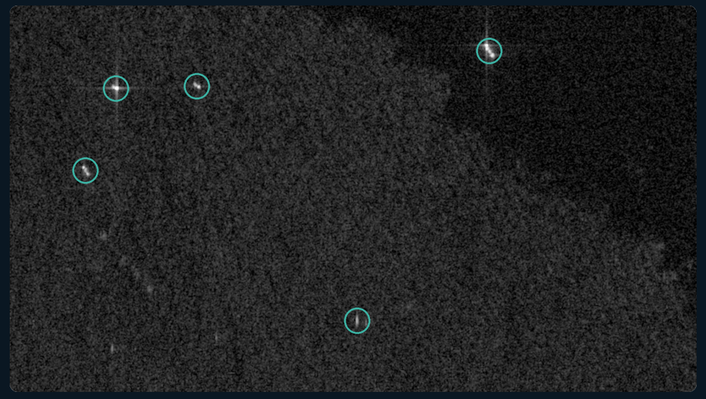
BUOY CAD MODEL

Team Interstellar · Tohraan Ahmad Khan · Ayush Khale · Aryan Khale · Mayank
BITS Pilani, Dubai Campus · November 2026

It already runs: real radar in, a checkable ship catalogue and a buoy model out

Detector

RUNNING

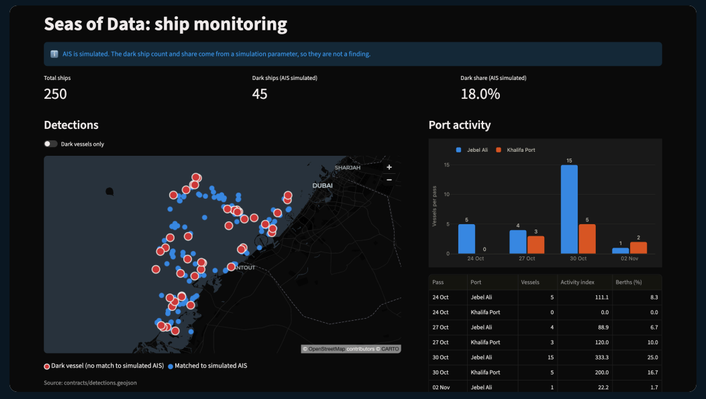


4
REAL PASSES

250
RADAR DETECTIONS

Dashboard

RUNNING

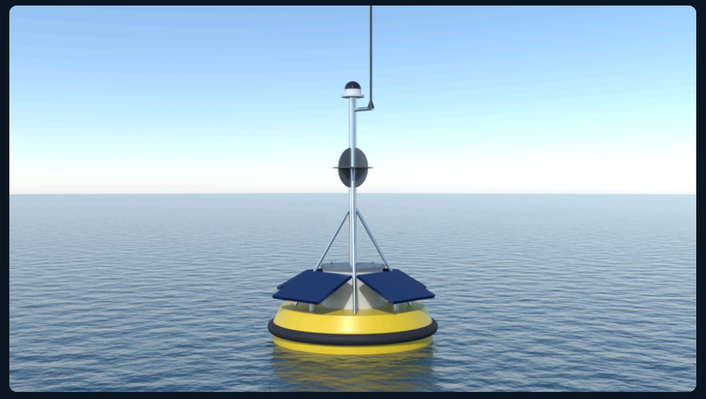


250
ON THE MAP

2
PORTS COUNTED

Buoy

CAD BUILT



14
CAD PARTS

4.5 m
TALL

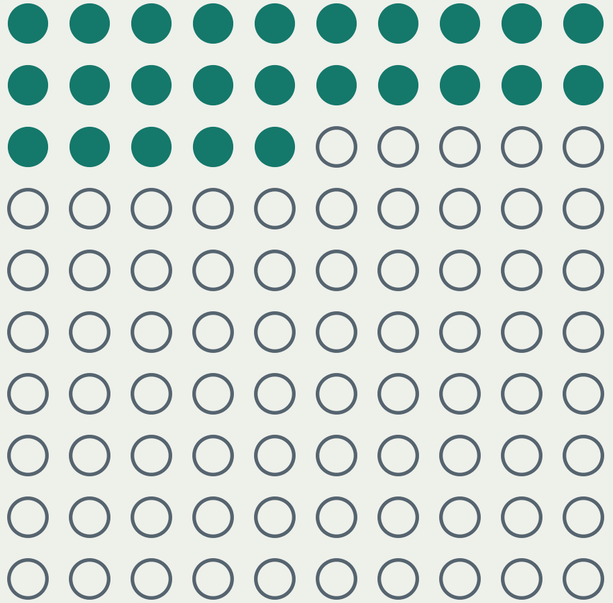
● REAL SATELLITE DATA ● REAL DETECTOR ○ AIS NOT JOINED YET ○ BUOY NOT YET BUILT

About three in four industrial fishing vessels are missing from public ship tracking

72–76%

of the world’s industrial fishing vessels are not publicly tracked.

Global figure, 2017 to 2021, not UAE-specific. Dark does not mean illegal.



Each circle = 1 of 100 vessels.
Filled = tracked.

SOURCE: PAOLO ET AL., NATURE 625, 85-91 (2024)

Sentinel-1 passes over Jebel Ali about every three days

COVERAGE

250 km swath, 10 m pixels, VV band

CONDITIONS

Day, night and through cloud

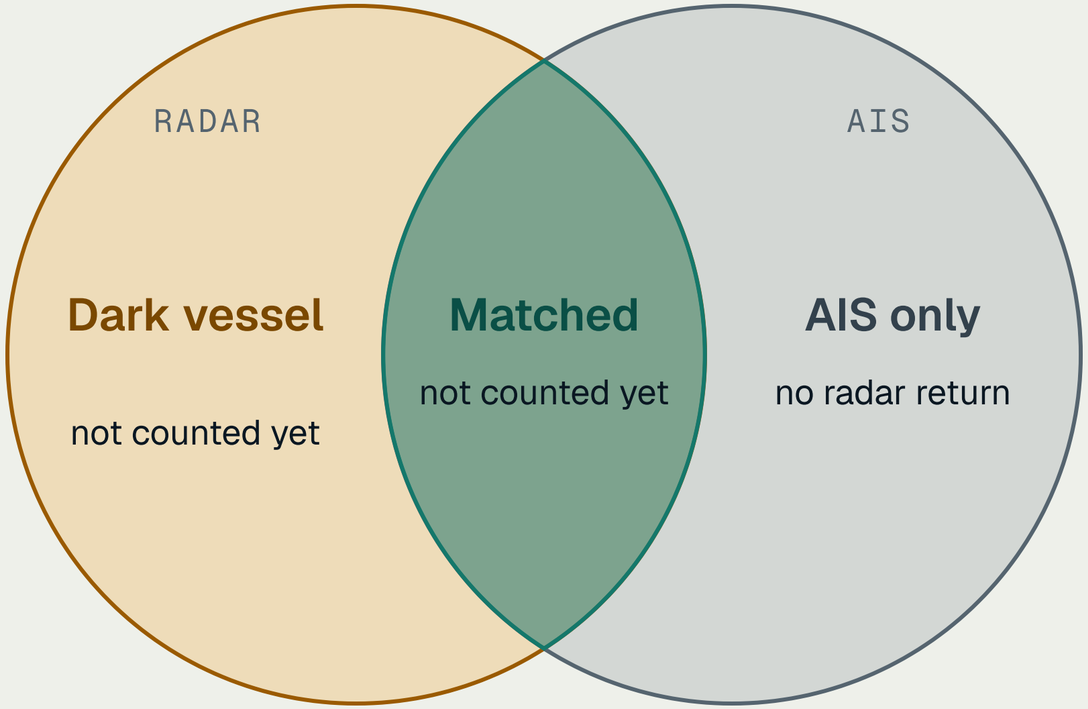
ACCESS

Free and open through Copernicus



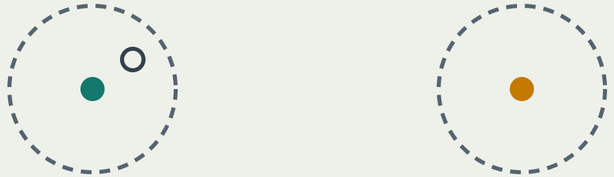
PASSES FOUND IN THE COPERNICUS CATALOGUE · 25.01°N 55.06°E

Radar minus AIS equals the dark vessels



MATCH GATE

500 m



MATCHED DARK

Dark is a lead for patrols, not a verdict.

AIS CHECK: NEXT STEP

One system: the satellite finds, the dashboard shows, the buoy confirms

01 · SPACE — REAL DATA

Sentinel-1 radar

Free, every three days, any weather

02 · PIPELINE — RUNNING

Detect and verify

Sea mask, CFAR, radar chips

03 · DASHBOARD — RUNNING

See and act

Map, chips, port counts

DARK-SHIP ALERT

AIS + PHOTO

04 · SEA — CAD BUILT

Solar buoy

AIS log, camera, oil sensor

THE LOOP — DESIGNED

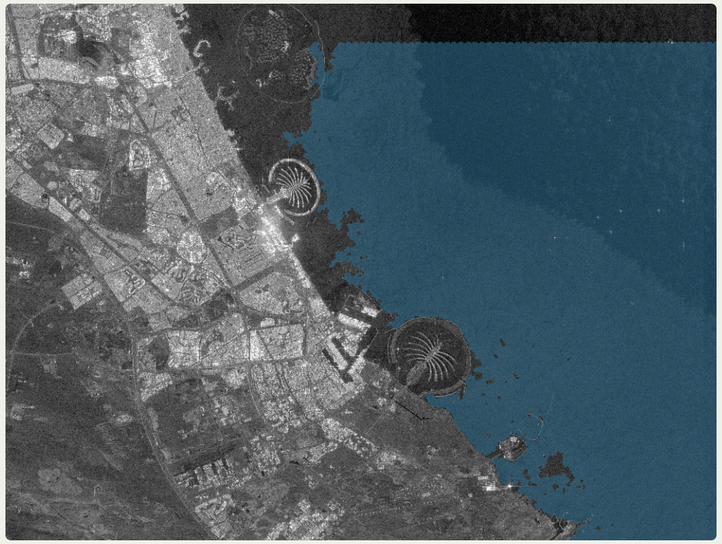
1. Radar flags a ship with no AIS report
2. The nearest buoy checks it with AIS and camera
3. Real AIS joins the catalogue

Real radar in, ships out, in three steps



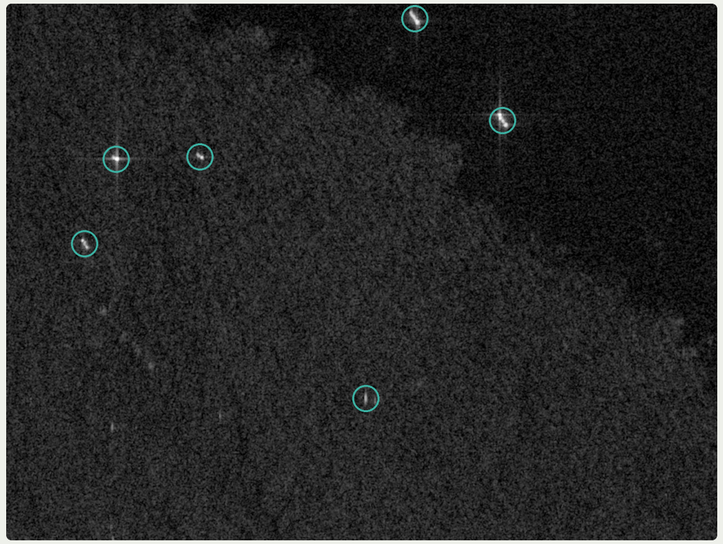
1 · RAW SCENE

2 Nov 2026. Bright = land and hulls.



2 · SEA MASK

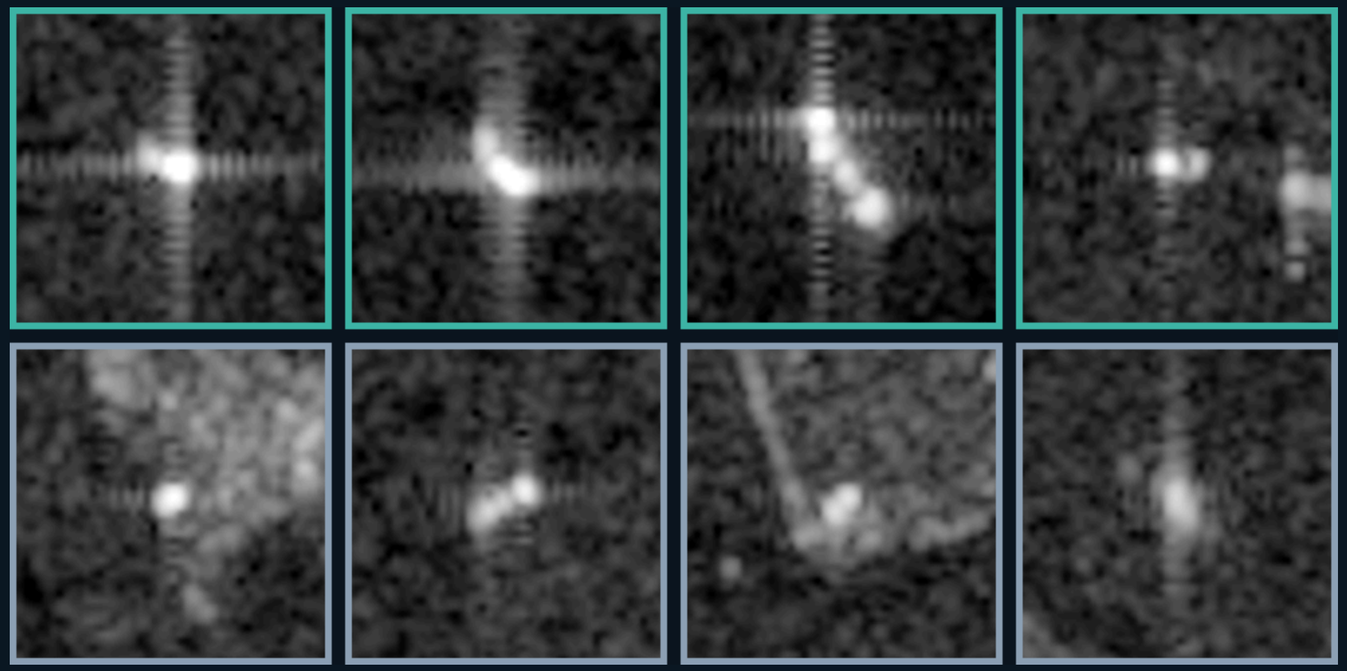
Blue = sea. Coast mapped once, reused.



3 · DETECTIONS

Teal ring = ship class. Slate = uncertain.

Close up: what the detector calls a ship, and what it is unsure about



- 480 m square around each target
- Top row: ship class. 94% look like ships on review
- Bottom row: uncertain. Only 9% do

Four real passes: 250 radar detections, 120 of them high-confidence ships



2 NOV 2026 · 02:14 UTC · 20 OF 73 DETECTIONS SHOWN

250
DETECTIONS

120
SHIP CLASS

121
UNCERTAIN

AIS NOT JOINED YET

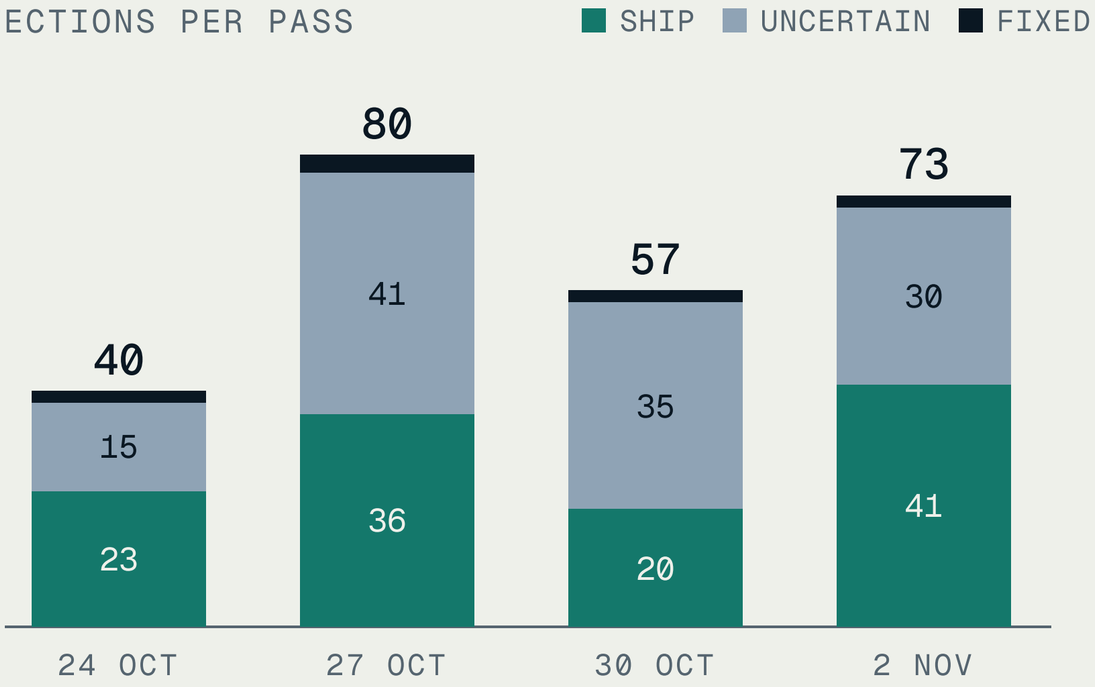
No ship is called dark here. 9 more repeat at the same spot in every pass.

HOW TO READ IT

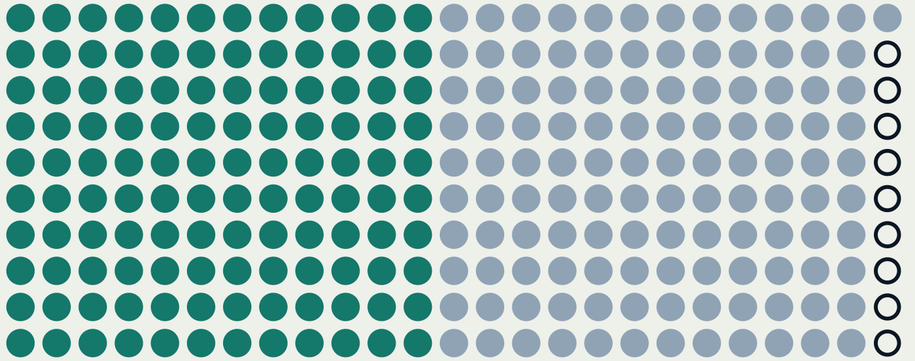
Teal ring: ship class. Slate ring: uncertain.

Each pass finds 40 to 80 radar targets; about half are high-confidence ships

DETECTIONS PER PASS



ALL 250 DETECTIONS, ONE DOT EACH



120 ship class **121 uncertain**

Radar only. No AIS joined, so nothing here is called dark.

First dashboard: every detection on a map, with the radar evidence to be added



VERSION 1

1 Map

Red and blue use simulated AIS: version 1 only.

2 Banner

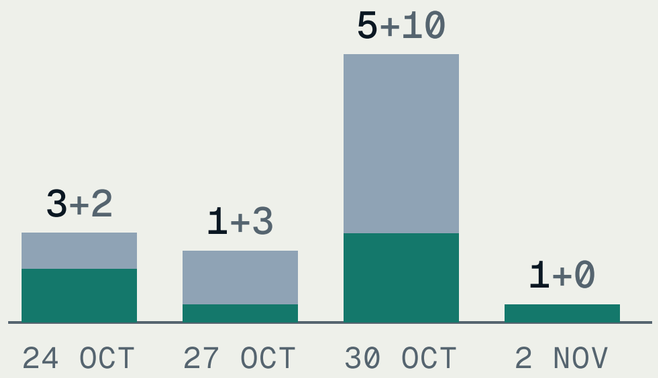
Says AIS is simulated.

3 Port panel

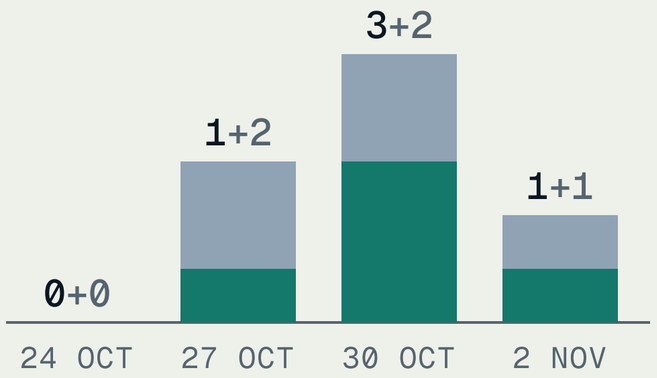
Detections per port and pass.

Near quays, 10 m radar is weakest: few confident ships, more doubtful ones

Jebel Ali



Khalifa Port



HOW TO READ

Teal: ship class. Slate: uncertain. Label: ship+uncertain.

WHY SMALL

Hulls beside quays look like buildings at 10 m.

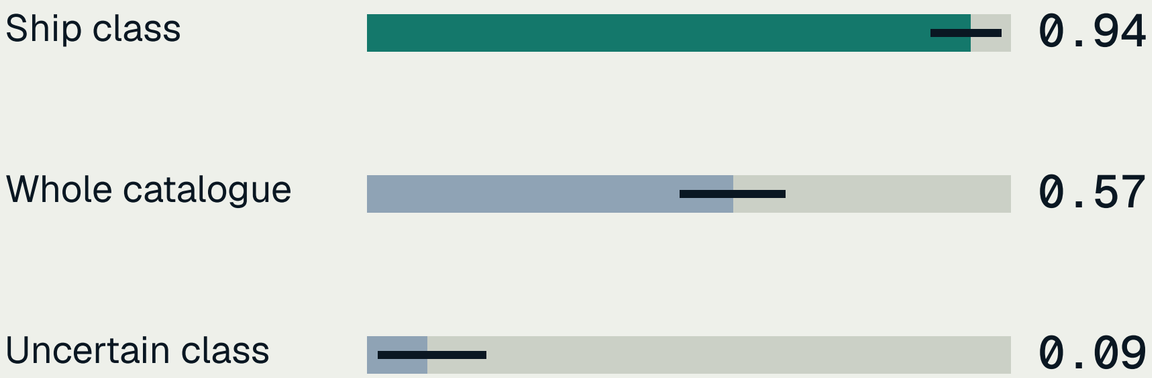
WHAT FILLS THE GAP

AIS plus a buoy camera, up close.

DETECTIONS PER PORT BOX · 4 PASSES · RADAR ONLY

On 120 radar chips we reviewed, 94% of the ship-class detections look like real ships

LOOK LIKE SHIPS · BAR = ESTIMATE · LINE = 95% RANGE



95% range: ship 0.88-0.99 · all 0.49-0.65 · uncertain 0.02-0.18

HOW WE MEASURED

120 random chips, equal draws per size band, judged by eye.

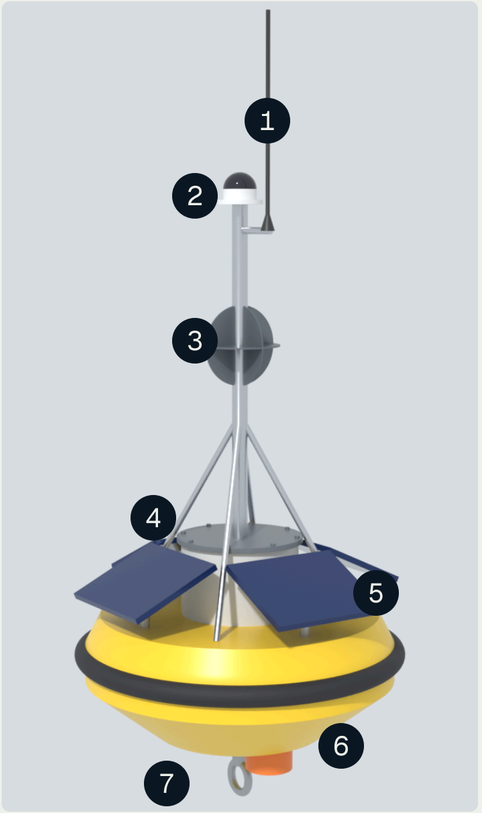
WHAT THIS IS NOT

No ground truth and no AIS. Recall is unknown. One reviewer so far.

BENCHMARK

On the HRSID benchmark: precision 0.91, recall 0.82. Finer pixels, so not Gulf accuracy.

A solar buoy checks what radar flags, between satellite passes



1 **AIS antenna**
Hears ships that broadcast

2 **Camera dome**
Sees silent ships

3 **Radar reflector**
Shows up on ships' radar

4 **Electronics bay**
Battery, processor, radio

5 **Four solar panels**
Power, tilted to shed spray

6 **Oil-sheen sensor**
Flags possible oil

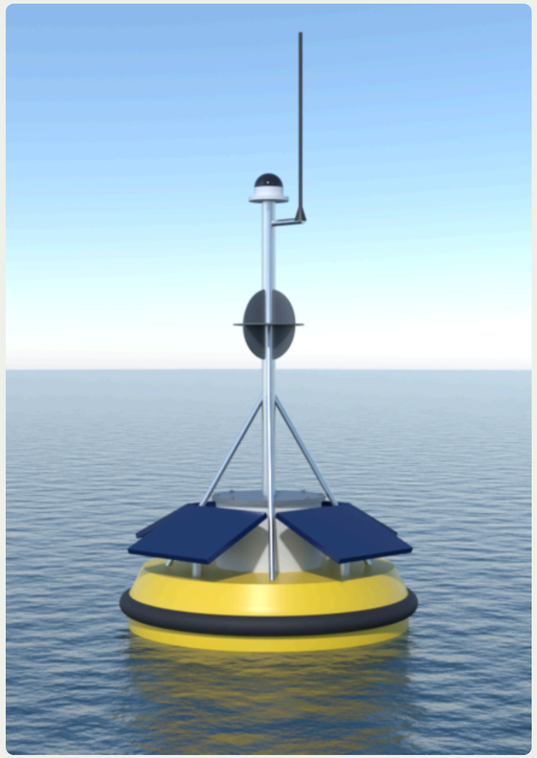
7 **Mooring eye**
Anchor chain point

CAD BUILT

4.5 m TALL	1.8 m HULL Ø
14 CAD PARTS	4 SOLAR PANELS

Design only: not built, not tested afloat.

One CAD model, four Blender views



3D VIEW

Hull, panels and mast together



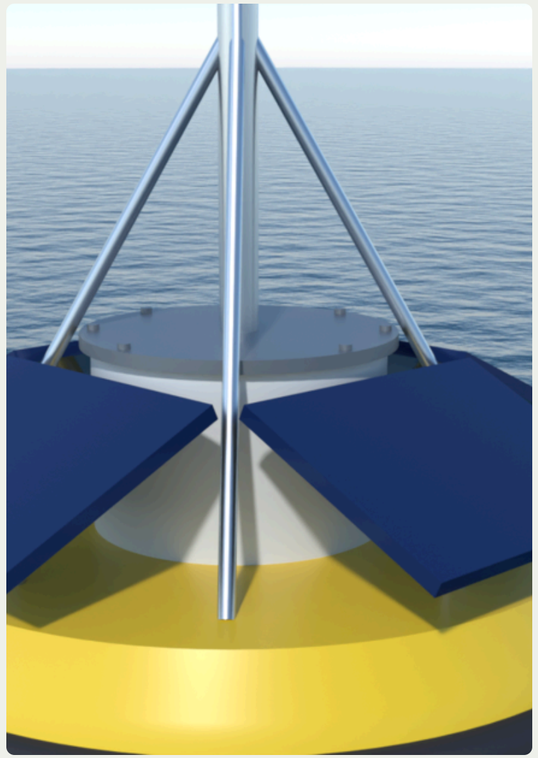
SIDE

Panels tilted 25°, sensor below



FROM ABOVE

Four panels, one per side

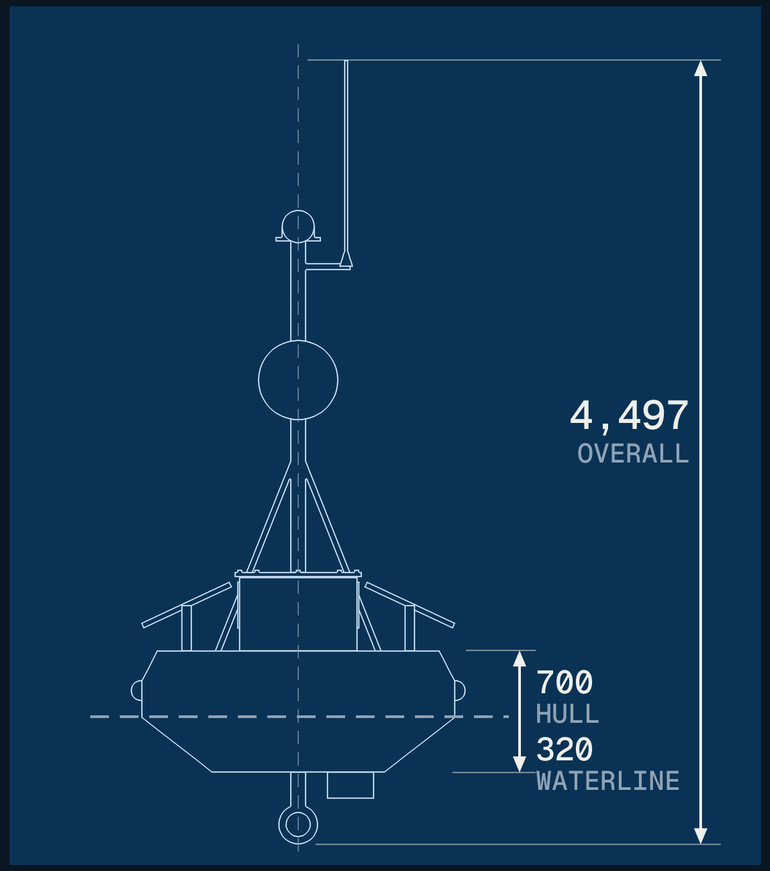


MAST HEAD

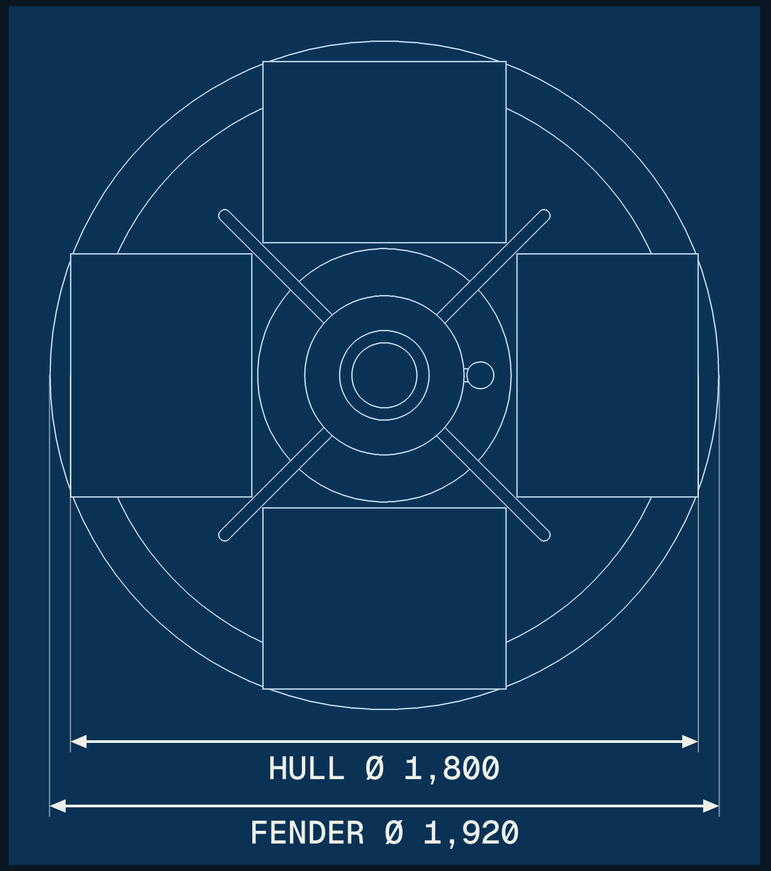
Camera, AIS and reflector

On paper: the buoy is 4.5 m tall and 1.8 m across

SIDE ELEVATION



PLAN VIEW



OVERALL HEIGHT	4,497 mm
HULL Ø	1,800 mm
HULL HEIGHT	700 mm
WATERLINE	320 mm
PANEL TILT	25°

DRAWN FROM THE CAD MESHES.
WATERLINE IS A DESIGN TARGET,
NOT A CALCULATION.

What is real, what is not joined yet, and what comes next

REAL AND RUNNING

- 4 Sentinel-1 passes, hashed
- 250 detections with radar chips
- Measured: 94% of ship class
- Buoy CAD model

NOT YET

- AIS joined: matcher built, feed pending
- Recall: no ground truth
- Buoy afloat: not built or tested

NEXT

- Real AIS check on new passes
- Retrain on xView3-SAR
- Add oil-spill detection

Data, models and licences

Component	Used for	Source and terms
Sentinel-1 GRD	Real Gulf scenes, VV band	ESA, via Copernicus Data Space, free and open
HRSID	Training and held-out test of the ship detector	Wei et al., IEEE Access; repository under GPL-3.0
YOLO11	Detector, fine-tuned from pretrained weights	Ultralytics
Port boxes	Counting ships inside Jebel Ali and Khalifa Port	Drawn by us, checked against OpenStreetMap extents

Contains modified Copernicus Sentinel data 2026.

Method and parameters

Step	Setting	Reason
Sea mask	Static lon/lat grid, 0.001°, built from a calm-sea pass	Windy sea is as bright as desert, so brightness cannot find the coast
Detector	CFAR, 61 px window, 8 sigma, at least 3 px, at most 450 m	Needs no training and works at 10 m pixels
Merge	Returns within 150 m become one ship	One hull often splits into several blobs
Geocoding	Ground control points in each GRD file	Checked against Palm Jumeirah and Palm Jebel Ali
AIS match	Not joined yet. Matcher: +/-10 min, gate 200 m + half hull + drift	Real feed needs a key; coverage must be proven before anything is called dark
YOLO11s	Fine-tuned on HRSID, scored on 1,962 held-out images: precision 0.91, recall 0.82, mAP50 0.90	Benchmark score only; Gulf passes use CFAR